CSE251: System Programming

HW #1

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Department of Computer Science and Engineering

HW #1 (100 points)

- Due March 24, 2026 : 11:59 pm

- Platform
  - We will work on Ubuntu 22.10 (x86-64).
  - The server we provide already runs **Ubuntu 22.10 (x86-64)**.
    - To access your assigned server, please read the notice **“Guide to SSH Access and Linux Basics”** on Blackboard.

- Responsible TA: Donggyu Kim (dgkim@unist.ac.kr)

HW #1

- Read a file that contains characters only “0”, “1” or “\n(newline)”
  E.g.,0101010101000001010100101111010101000100101010111010101 0
  - The input does NOT have any other character.
- File name that your program needs to read is **<u>“input.txt”</u>**
  (e.g., `fopen(“input.txt”, r);`)
  - The input does not have any other characters such as newline.
- Store the data in the memory in each “bit” and represent the data in the file using various data types in C (with following little endian manner)
- You can use ‘`malloc`’ and ‘`strlen`’ to read lines of the input file
  - The lengths of each input line is not fixed, but it is aligned by 64 characters.
- Don’t assume the size of the input file (it can be a huge file)
  - Considering unexpected input large size memory, please **MUST** use ‘`realloc`’

Requirements

- You MUST provide your Makefile.
  - We will compile your code using your Makefile with the ‘`make`’ command.
- The assignment MUST be compilable.
  - If the compilation is failed, we can not assure the scoring.
- You MUST provide at least one ‘.c’ file and one ‘.h’ file.
- The compiled program MUST be named as ‘hw1’ (use –o option).
- You MUST provide simple and proper comments on your code.
- You MUST print the data through the file (i.e., output.txt).

Requirements

- Your code MUST prints out the data as decimal numbers
  - When you print the data using float or double types, you MUST print out 4 digits after the decimal separator (e.g., `12.3456`).
  - When you print the data using float or double types, you MUST print out 4 digits after the decimal, filled with 0s (e.g., `12.0000`), even if there are no decimal places.
- You MUST print out ASCII character (if possible) as well.
  - You MUST print its ASCII code using a dot "." if there is no ASCII character corresponding the number.
- The printing format MUST follow the example's one (please see the next slide's example).
- You can test yourself by using '`random_hex_generator`' binary.

Requirements

- You MUST read and print the input data by using the following data types (with the *exactly same order*).
  - Signed char
  - ASCII codes
  - Unsigned char
  - Signed int
  - Unsigned int
  - Signed float
  - Signed double

Requirements

[input.txt]

- Multiple lines of bit sequences (See the attached ‘input.txt’)

```
010000000110001101100...010110010100010010000 1
010000000110001101100...1111010101011001010001001000 01
```

[output.txt]

- You MUST follow the printing format (See the attached ‘output.txt’)

```
Ubuntu
[*] Signed Char: 33 81 86 61 109 100 99 64 33 81 86 61 109 100 99 64 33 81 86 61 109 100 99 64 33 81 86 61 109 100 99 64
 33 81 86 61 109 100 99 64 33 81 86 61 109 100 99 64 33 81 86 61 109 100 99 64 33 81 86 61 109 100 99 64 33 81 86 61 109
 100 99 64 33 81 86 61 109 100 99 64
[*] ASCII Codes: ! Q V = m d c @ ! Q V = m d c @ ! Q V = m d c @ ! Q V = m d c @ ! Q V = m d c @ ! Q V = m d c @ ! Q V =
 m d c @ ! Q V = m d c @ ! Q V = m d c @ ! Q V = m d c @
[*] Unsigned Char: 33 81 86 61 109 100 99 64 33 81 86 61 109 100 99 64 33 81 86 61 109 100 99 64 33 81 86 61 109 100 99
64 33 81 86 61 109 100 99 64 33 81 86 61 109 100 99 64 33 81 86 61 109 100 99 64 33 81 86 61 109 100 99 64 33 81 86 61 1
09 100 99 64 33 81 86 61 109 100 99 64
[*] Signed Int: 1029067041 1080255597 1029067041 1080255597 1029067041 1080255597 1029067041 1080255597 1029067041 10802
55597 1029067041 1080255597 1029067041 1080255597 1029067041 1080255597 1029067041 1080255597 1029067041 1080255597
[*] Unsigned Int: 1029067041 1080255597 1029067041 1080255597 1029067041 1080255597 1029067041 1080255597 1029067041 108
0255597 1029067041 1080255597 1029067041 1080255597 1029067041 1080255597 1029067041 1080255597 1029067041 1080255597
[*] Signed Float: 0.0523 3.5530 0.0523 3.5530 0.0523 3.5530 0.0523 3.5530 0.0523 3.5530 0.0523 3.5530 0.0523 3.5530 0.05
23 3.5530 0.0523 3.5530 0.0523 3.5530
[*] Signed Double: 155.1383 155.1383 155.1383 155.1383 155.1383 155.1383 155.1383 155.1383 155.1383 155.1383
~
~
```

Submission

- You should submit your code with a code description that explains your code (i.e., comments in the file). In the description, your code must be well commented to explain your algorithm. Make your code .zip file with “StudentID_YourName_HW1.zip” and submit your .zip file on blackboard.
- ex) 20265032_DonggyuKim_HW1.zip
  - hw1.c // hw1 code
  - hw1.h // hw1 header file
  - Makefile
  - report.pdf // hw1 code description
- Please note that Problem’s input and output file names are ‘input.txt’ and ‘output.txt’.
- For the report, please write it briefly.
  More specifically, it is enough to copy and paste the important part of your source code and briefly describe it.

Grading Criteria

- We will grade strictly based on (1) the accuracy of the output relative to the input and (2) the quality of the summary and organization of the report.pdf file.

- Note that we will evaluate your code in our server environment. If your code does not run correctly in our environment, you will receive zero points for this homework. No exceptions will be made.

Questions

- If you have any requests or questions (technical difficulties, late submission due to inevitable circumstances, etc.), please ask the TAs on Blackboard.
- We generally encourage the use of Blackboard for discussions. However, for urgent issues or secret issues, you can send an email to the responsible TA, Donggyu Kim (dgkim@unist.ac.kr).

Be Careful about Plagiarism!

- We will conduct strict cross-plagiarism detection during evaluation, including comparisons against answers generated by ChatGPT, code found online, and other students' submissions. Therefore, we strongly discourage consulting ChatGPT or online solutions.

- Last semester, we identified several cases of plagiarism using an automated detection tool. If you are found to be engaged in **deep collaboration** with other students, both the provider and the recipient will receive penalties. In the worst case, this may result in receiving an **F** for the course.